

RHEED-conditioned AFM generation across measured roughness
fixed ordering, shared ± 8.3 nm color scale

RHEED key-frame ROI

Generated AFM (nm)

Measured AFM (nm)

6029 [strict LOO]
confidence 100/100

Rq 2.58 nm
90% PI [0.86, 9.64]

Rq 2.56 nm

6070 [strict LOO]
confidence 0/100

Rq 2.43 nm
90% PI [0.79, 9.26]

Rq 3.05 nm

6062 [strict LOO]
confidence 63/100

Rq 2.93 nm
90% PI [0.85, 9.74]

Rq 3.09 nm

6085 [strict LOO]
confidence 99/100

Rq 3.09 nm
90% PI [0.85, 9.89]

Rq 3.13 nm

6056 [validation]
confidence 61/100

Rq 3.06 nm
90% PI [0.87, 9.94]

Rq 3.23 nm

6047 [strict LOO]
confidence 83/100

Rq 2.37 nm
90% PI [0.80, 9.28]

Rq 3.07 nm

Training rows are strict leave-one-growth-group-out predictions; validation rows use the frozen training fit. Confidence is a relative interval-width index, not a probability.